

STORE SALES ANALYSIS

Retail Sales, Customer & Product Performance Analytics

Tools: Power BI • Power Query • M / DAX-enabled semantic modeling.

1. Executive Overview

The project analyzes a retail store's sales performance, customer purchasing behavior, product trends, profitability, promotions, and geographic distribution. The objective was to transform raw relational data into a structured analytical model and deliver an interactive Power BI report for business decision-making.

Business questions addressed

- Which products generate the highest and lowest sales?
- How are sales changing over time?
- How are profit and net sales related?
- Which promotion categories have the largest discount impact?
- Which cities contribute most strongly to the customer/sales footprint?
- Are high-volume categories necessarily the most profitable?
- Can high-value / 'power user' customers be identified for targeted action?

Project value

The strongest part of the project is the end-to-end workflow: data integrity restoration → business-rule calculations → dimensional modeling → analytical visuals → business recommendations.

2. Dataset & Data Preparation

Relational dataset structure

Table	Business role	Important fields / purpose
Fact Sales	Transactional fact table	Units sold, order date, revenue-related fields
Dim Product	Product master	Product ID, Product Name, Unit Price
Dim Promotion	Promotion master	Promotion Name, discount information
Dim Customers	Customer master	Customer details and City

Power Query / ETL work

- Data imputation via merging: Left Outer Joins brought Price and Discount % into the Fact table and filled the reported null values.
- Business calculations: M-code logic calculated Total Sales as Units × Price and Discount Value as Sales × Discount %.
- Schema cleanup: Data types were standardized; ID columns were converted to Text where appropriate.
- Data profiling: Column Profiling was used to inspect completeness and quality before analysis.
- Transaction tracking: An Index Column was added to provide a unique transaction/order identifier for reporting.

3. Data Modeling

The model follows a star-schema design: the Fact Sales table acts as the central transactional table, while Product, Promotion, and Customer dimensions provide descriptive business context.

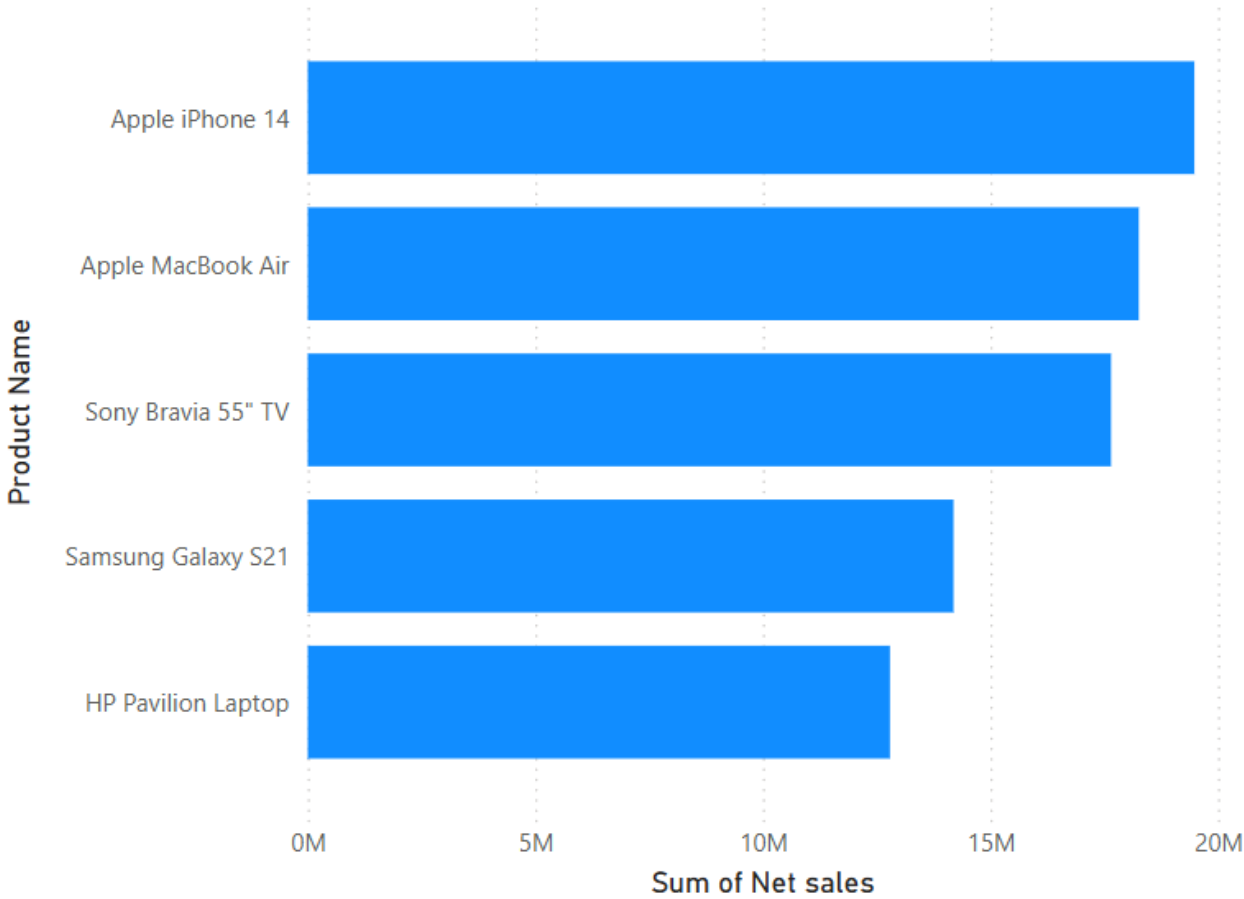
Modeling decisions

- Star schema: separates transactional measures from reusable descriptive dimensions.
- Two date tables: created using CALENDARAUTO() to support separate date-analysis paths.
- Active/inactive date relationships: one active and one inactive relationship were created to support period-over-period analytical scenarios.
- Business-ready measures/fields: the model exposes sales, profit, discount, product, promotion, customer, and date dimensions for interactive analysis.

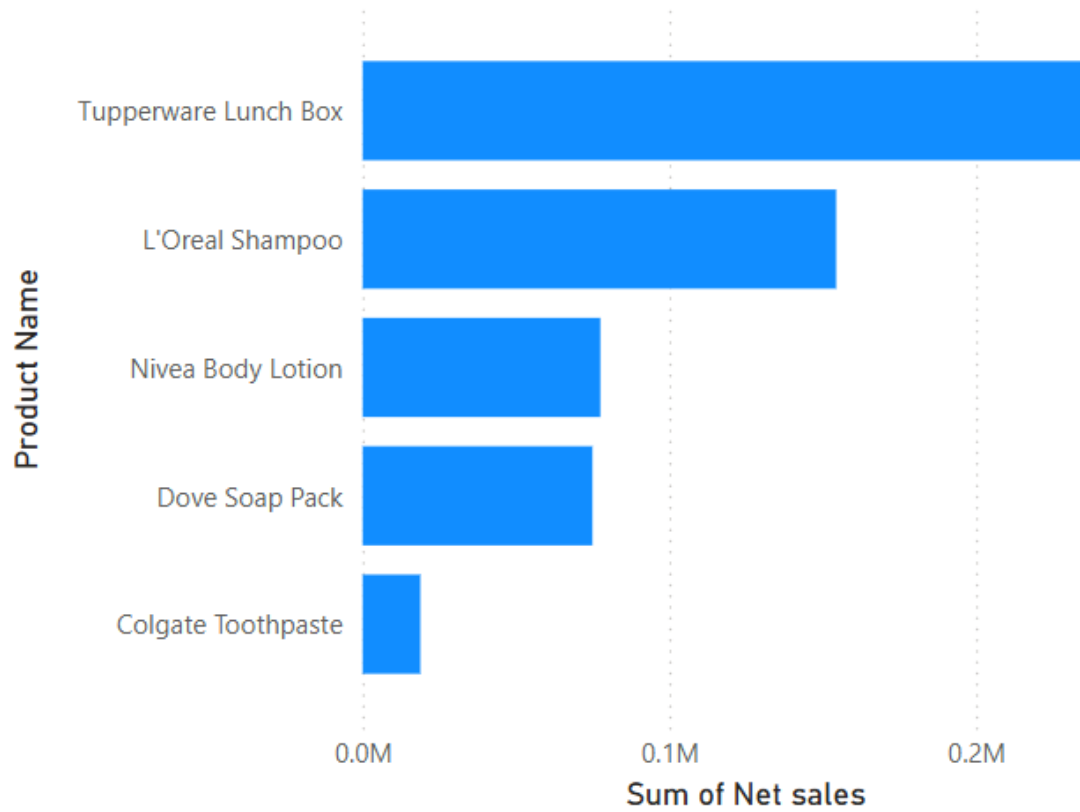
4. Power BI Dashboard

Dashboard Page 1 — Product Performance

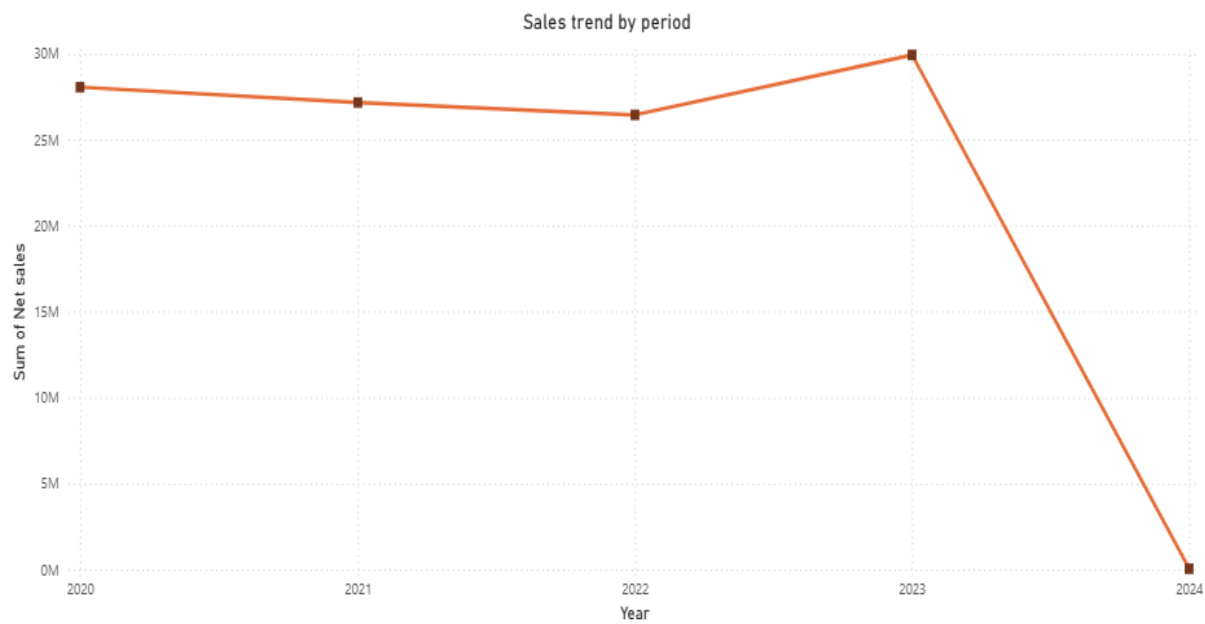
Top 5 products by Sales



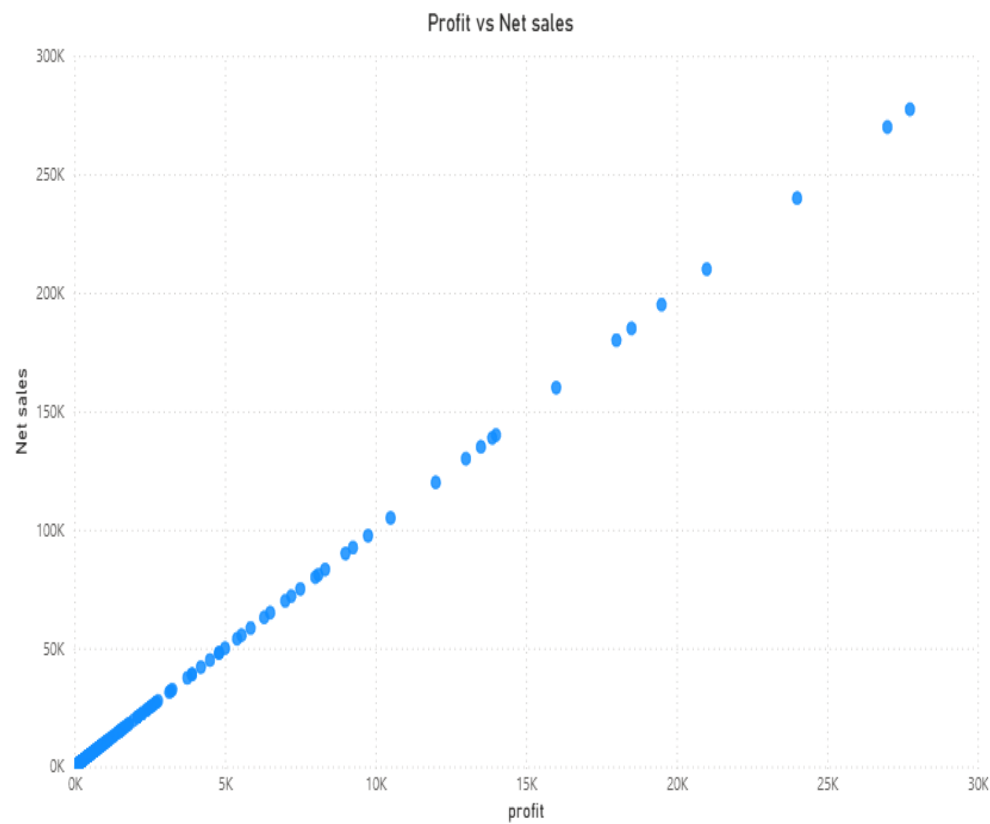
Bottom 5 products by sales



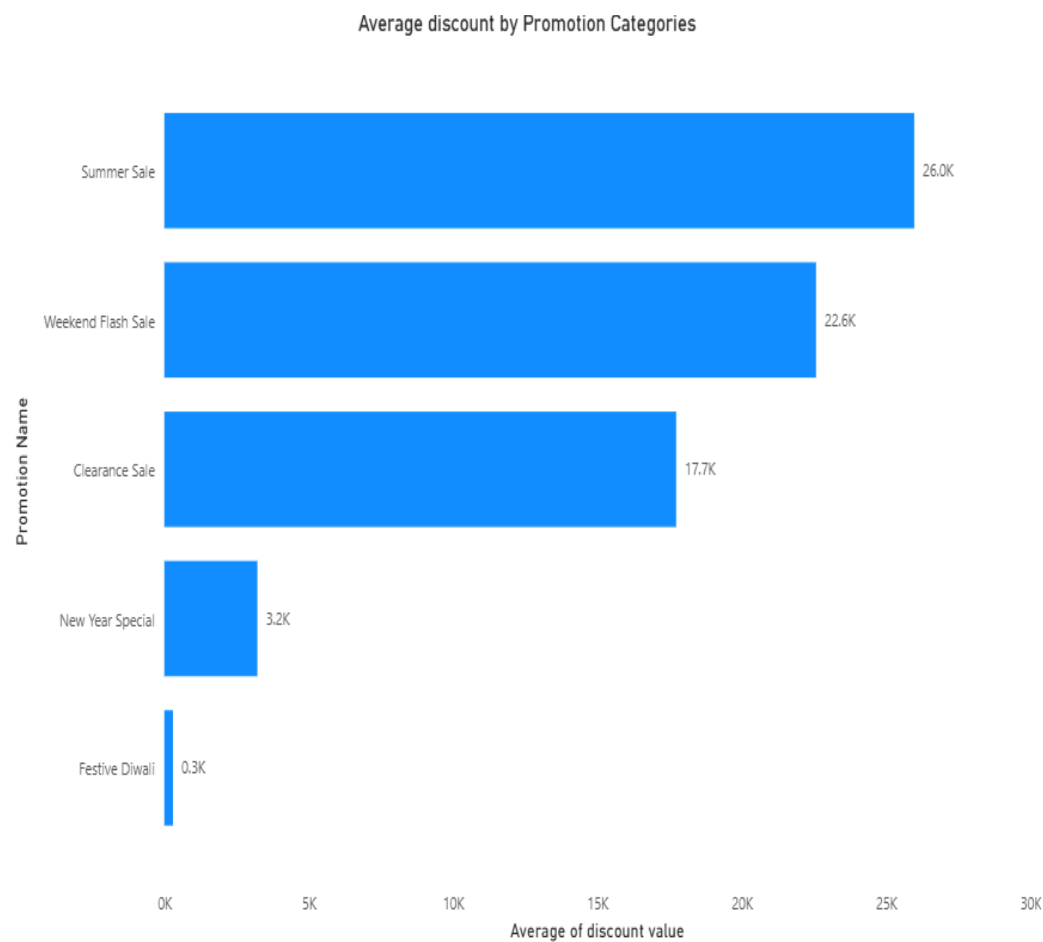
Dashboard Page 2 — Sales Trend

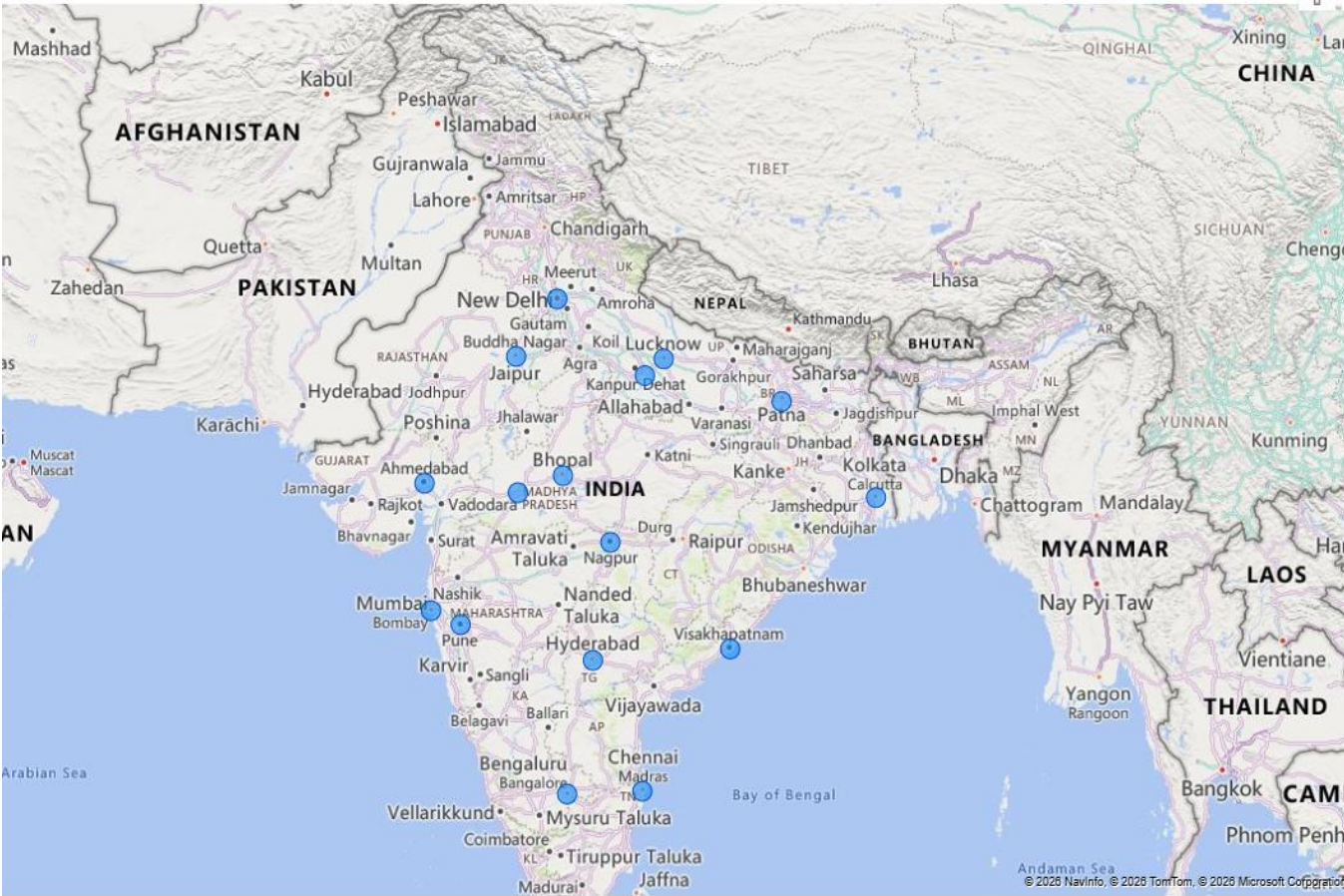


Dashboard Page 3 — Profitability Analysis



Dashboard Page 4 — Promotion & Discount Analysis





5. Business Findings

Product performance

- Apple iPhone 14 was identified as the top-selling product, while Colgate toothpaste ranked at the bottom.
- The top/bottom product views provide a direct way to prioritize inventory, merchandising, and product-level investigation.

Sales trend

- Net sales declined from 2020 through 2022, increased through 2023, and then declined sharply again in 2024.
- This pattern supports deeper investigation into product mix, promotions, customer behavior, and regional performance.

Promotion performance

- Discount value was highest during the Summer Sale and lowest during Diwali in the project's analysis.

Customer and geography

- Delhi was identified as the primary revenue-driving city.
- The analysis also identified a group of high-value 'Power Users' contributing a disproportionate share of revenue.

Profitability

- Clothing and Electronics generated high transaction volumes, but volume did not automatically translate into the strongest profitability.
- Accessories were identified as an example of a lower-overhead sub-category with a stronger profit-per-unit relationship despite lower total sales.

6. Recommendations

1. Cross-selling analysis

Perform Frequently Bought Together analysis to determine whether customers in major cities such as Delhi buy recurring combinations of Electronics and Accessories. This could support bundles and cross-sell campaigns.

2. Sales forecasting

Add a trend line or forecasting model in the next dashboard version to estimate the next three months of sales using historical patterns.

3. Regional marketing

Maintain Delhi's position with localized campaigns while testing targeted expansion discounts in lower-performing cities such as Kanpur.

4. Profitability-led merchandising

Avoid optimizing purely for transaction volume. Prioritize products/sub-categories that combine acceptable sales volume with stronger profit-per-unit economics.

5. Customer segmentation

Use the identified Power User segment for retention and personalized offers, while separately targeting lower-value customers with conversion and basket-size initiatives.